Repeated-Measures ANOVA

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Introduction

Repeated-measures ANOVA (rmANOVA) handles designs where the same subjects are measured at multiple time points or conditions. By treating subject as a blocking factor, it controls between-subject variability and gains power over between-subjects analysis. The main complication is the **sphericity** assumption on the covariance of the repeated measures.

Prerequisites

One-way ANOVA, dependence structure in repeated measures.

Theory

For one within-subjects factor with k levels measured on n subjects, the model is

$$Y_{ij} = \mu + \alpha_i + \pi_j + \varepsilon_{ij},$$

where π_j is a random subject effect and α_i is the fixed effect of level i.

Sphericity assumes the variances of all pairwise differences between repeated measurements are equal. Mauchly's test assesses this assumption.

If sphericity is violated:

- **Greenhouse-Geisser correction**: adjusts degrees of freedom by multiplying by $\hat{\varepsilon}_{\text{GG}} \in [1/(k-1), 1]$.
- **Huynh-Feldt correction**: less conservative for mild violations.

Modern alternative: fit a linear mixed-effects model (`lmer`) with an unstructured or compound-symmetric covariance – more flexible, handles missing data.

Assumptions

- Normality of within-subject residuals.
- Sphericity (or apply a correction).
- Subjects are independent; measurements within a subject are not.

R Implementation

```r
library(afex); library(emmeans); library(rstatix)
set.seed(2026)

# 20 subjects measured at 4 time points
n_subj <- 20
times  <- 4
subject <- rep(1:n_subj, each = times)
time    <- rep(1:times, n_subj)
```

```r
# Simulate with a time effect and subject-specific intercepts
df <- data.frame(
  subject = factor(subject),
  time    = factor(time),
  y       = rnorm(n_subj * times, mean = 50 + 2 * time, sd = 6) +
            rep(rnorm(n_subj, 0, 4), each = times)
)

# Using afex::aov_car
fit <- aov_car(y ~ time + Error(subject/time), data = df)
fit

# rstatix pipeline
df |> anova_test(dv = y, wid = subject, within = time)

# Post-hoc with Bonferroni
emm <- emmeans(fit, ~ time)
pairs(emm, adjust = "bonferroni")
```

Output & Results

```
Anova Table (Type 3 tests)

Response: y
   Effect         df  MSE      F  ges  p.value
1    time  2.61, 49.5 28.7  12.39  0.19  <.001 ***

Mauchly Tests for Sphericity:
 Effect   W      p
  time  0.87  0.78   # sphericity holds

Greenhouse-Geisser eps: 0.87
```

Large within-subject effect of time ($F(2.61, 49.5) = 12.39$, $p < 0.001$, generalised $\eta^2 = 0.19$); sphericity holds so no correction needed.

Interpretation

"A one-way repeated-measures ANOVA revealed a significant effect of time ($F(2.61, 49.5) = 12.39$, $p < 0.001$, generalised $\eta^2 = 0.19$; Greenhouse-Geisser correction applied). Bonferroni post-hoc comparisons identified significant increases from time 1 to time 3 and from time 2 to time 4."

Practical Tips

- Check Mauchly's test; when rejected, apply Greenhouse-Geisser or Huynh-Feldt.
- For complex designs (multiple within factors, between x within interactions), a linear mixed model is preferred.
- Missing data in rmANOVA causes the entire subject to be dropped (listwise); mixed models handle it gracefully.
- Report generalised η^2 for repeated-measures designs (partial η^2 is upwardly biased).
- For non-normal residuals, use Friedman or an aligned-rank transform (`ARTool::art`).

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests